

D798

Emerging Computing Systems

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A global shopping-cart database loses the link between its US and EU data centers. The business wants customers to keep adding items even if carts briefly disagree and are merged later. The database runs on 400 nodes. Which choice fits?

- A. Favor consistency, rejecting writes until the link returns.
- B. Keep full consistency and full availability during the split.
- C. Favor availability, accepting temporary inconsistency that is reconciled later.
- D. Add more nodes to avoid the trade-off.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A bank ledger must never show two different balances for one account, so during a network partition it rejects some writes. Which CAP trade-off is this?

- A. Eventual consistency.
- B. Consistency over availability (CP).
- C. Availability over consistency (AP).
- D. CA, since partitions do not affect it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A factory robot's control loop needs a reply within 8 ms. The nearest cloud region is 1,000 km away, and signals in fiber travel about 200,000 km per second. The factory runs three shifts. Can the cloud region meet the deadline?

- A. No; travel alone takes 10 ms round trip, so process at the edge.
- B. It depends on how many shifts run.
- C. Yes; travel takes only 5 ms.
- D. Yes, with a faster cloud server.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Signals in fiber travel about 200,000 km per second. How far away can a server be if round-trip travel must take no more than 4 ms?

- A.** About 100 km
- B.** About 400 km
- C.** About 200 km
- D.** About 800 km

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A data center draws 1,500 kW in total, and its IT equipment uses 1,000 kW. It was built in 2019 beside a river. What is its PUE?

- A.** It depends on the river.
- B.** 0.5
- C.** About 0.67
- D.** 1.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A site's IT load is 2,000 kW and its PUE is 1.25. What is its total facility power?

- A.** 2,000 kW
- B.** 500 kW
- C.** 1,600 kW
- D.** 2,500 kW

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A ticket site gets almost no traffic most days but huge spikes for a few minutes when concerts go on sale. The team has three developers who all use a popular code editor. Which model best fits cost and scaling?

- A.** Serverless functions that scale automatically and bill per use, planning for cold starts.
- B.** One very large virtual machine.
- C.** Standardize on a different code editor.
- D.** Reserved servers sized for peak, running all year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A team wants to run 50 small Linux services on one host with fast startup and low overhead. The services were written by four different teams. Which approach fits best?

- A. Containers sharing the host's kernel.
- B. A separate virtual machine for each service.
- C. A separate physical server for each service.
- D. One VM per writing team.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A provider must run untrusted code from many customers, some needing different operating systems, with the strongest isolation. Which approach fits?

- A. Separate processes on one operating system.
- B. Containers, since they always isolate as well as VMs.
- C. Virtual machines, each with its own kernel.
- D. Containers on one shared kernel.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A company's VPN lets any connected laptop reach every internal app. When one laptop was compromised, the intruder reached dozens of systems. The VPN vendor is top-rated. Which model addresses this?

- A. Switch to another VPN vendor.
- B. A stronger perimeter firewall.
- C. Zero trust: verify identity, device health, and permission for each request, with least privilege.
- D. Trust internal traffic even more to reduce friction.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A company runs the same neural-network inference billions of times a day, and general-purpose CPUs use too much energy per inference. The company has offices in five countries. Which approach reflects current trends?

- A.** One general CPU type is always best for every workload.
- B.** Specialized accelerators such as GPUs or neural processors built for that math.
- C.** CPUs with much higher clock speeds.
- D.** One data center per office country.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A CIO reads that quantum computers will soon make all business software faster. The company's IT budget is \$40 million. Which statement is accurate?

- A.** Quantum computing has no effect on security.
- B.** All software will run faster on quantum computers.
- C.** The impact depends on the size of the IT budget.
- D.** Quantum helps only certain problems, but it threatens today's public-key encryption, so plan for post-quantum cryptography.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A global shopping-cart database loses the link between its US and EU data centers. The business wants customers to keep adding items even if carts briefly disagree and are merged later. ~~The database runs on 400 nodes.~~ Which choice fits?

Target word: CAP choice **Decides:** Business prefers always answering.

A	Inverted operation	Opposite of the stated business priority.
B	Absolute language	A partition forces a choice between them.
C	Correct	An AP choice keeps accepting writes during the partition.
D	Camouflage	Node count does not remove the partition choice.

2. Answer B Concept 1 • transfer

A bank ledger must never show two different balances for one account, so during a network partition it rejects some writes. Which CAP trade-off is this?

Target word: name the trade-off **Decides:** Consistency chosen over availability.

A	Adjacent category	That is what AP systems use, not this ledger.
B	Correct	Rejecting writes keeps one true balance.
C	Inverted operation	AP would keep accepting writes.
D	Confusion set	Any distributed system must handle partitions.

3. Answer A Concept 2

A factory robot's control loop needs a reply within 8 ms. The nearest cloud region is 1,000 km away, and signals in fiber travel about 200,000 km per second. ~~The factory runs three shifts.~~ Can the cloud region meet the deadline?

Target word: latency check **Decides:** Round trip is 2,000 km.

A	Correct	$2,000 \text{ km} / 200,000 \text{ km/s} = 0.01 \text{ s} = 10 \text{ ms}$.
B	Camouflage	Shifts do not change physics.
C	Dropped element	Counts one way and drops the return trip.
D	Adjacent category	Server speed cannot cut travel time.

4. Answer B Concept 2 • transfer

Signals in fiber travel about 200,000 km per second. How far away can a server be if round-trip travel must take no more than 4 ms?

Target word: solve for distance **Decides:** 2 ms each way.

A	Inverted operation	Divides by 4 again instead of using the one-way time.
B	Correct	$0.002 \text{ s} \times 200,000 \text{ km/s} = 400 \text{ km}$.
C	Confusion set	Halves the distance a second time.
D	Dropped element	Uses the full 4 ms as one-way time.

5. Answer D Concept 3

A data center draws 1,500 kW in total, and its IT equipment uses 1,000 kW. ~~It was built in 2019 beside a river.~~ What is its PUE?

Target word: PUE **Decides:** Total ÷ IT.

A	Camouflage	Location may affect cooling, but PUE comes from the two numbers.
B	Confusion set	Overhead ÷ IT is not PUE.
C	Inverted operation	IT ÷ total flips the ratio.
D	Correct	1,500 / 1,000 = 1.5.

6. Answer D Concept 3 · transfer

A site's IT load is 2,000 kW and its PUE is 1.25. What is its total facility power?

Target word: solve for total power **Decides:** Total = PUE × IT.

A	Wrong stem number	The IT load alone.
B	Dropped element	Only the overhead, dropping the IT load.
C	Inverted operation	Divides instead of multiplying.
D	Correct	1.25 × 2,000 = 2,500.

7. Answer A Concept 4

A ticket site gets almost no traffic most days but huge spikes for a few minutes when concerts go on sale. The team has three developers ~~who all use a popular code editor.~~ Which model best fits cost and scaling?

Target word: compute model **Decides:** Idle most of the time, short large spikes.

A	Correct	Pays nothing when idle and scales for spikes.
B	Adjacent category	Cannot scale out for spikes and still costs when idle.
C	Camouflage	Tools do not decide hosting.
D	Inverted operation	Pays for peak capacity that sits idle.

8. Answer A Concept 5

A team wants to run 50 small Linux services on one host with fast startup and low overhead. ~~The services were written by four different teams.~~ Which approach fits best?

Target word: packaging choice **Decides:** Density and fast startup on one OS.

A	Correct	Light, fast, and dense for same-OS services.
B	Confusion set	Each VM carries a full OS, adding overhead.
C	Adjacent category	Far more hardware than needed.
D	Camouflage	Team count does not decide the platform.

9. Answer C Concept 5 • transfer

A provider must run untrusted code from many customers, some needing different operating systems, with the strongest isolation. Which approach fits?

Target word: isolation choice **Decides:** Untrusted tenants and mixed OS.

A	Adjacent category	Even weaker isolation.
B	Absolute language	Containers trade isolation for lightness.
C	Correct	A hypervisor boundary isolates more strongly and allows different OSs.
D	Inverted operation	Shared kernels are a weaker boundary.

10. Answer C Concept 6

A company's VPN lets any connected laptop reach every internal app. When one laptop was compromised, the intruder reached dozens of systems. ~~The VPN vendor is top-rated.~~ Which model addresses this?

Target word: security model **Decides:** Inside location granted broad trust.

A	Camouflage	The design, not the vendor, allowed broad access.
B	Adjacent category	The intruder was already inside the perimeter.
C	Correct	Removes blanket trust based on being on the network.
D	Inverted operation	Widens the exposure.

11. Answer B Concept 7

A company runs the same neural-network inference billions of times a day, and general-purpose CPUs use too much energy per inference. ~~The company has offices in five countries.~~ Which approach reflects current trends?

Target word: hardware strategy **Decides:** One heavy, repeated workload; energy matters.

A	Absolute language	Specialization is the current path to efficiency.
B	Correct	Specialized hardware gives far better performance per watt.
C	Confusion set	The power wall limits clock gains.
D	Camouflage	Office count does not decide hardware efficiency.

12. Answer D Concept 8

A CIO reads that quantum computers will soon make all business software faster. ~~The company's IT budget is \$40 million.~~ Which statement is accurate?

Target word: realistic view of quantum **Decides:** Speedups are problem-specific; cryptography is at risk.

A	Inverted operation	It endangers widely used public-key methods.
B	Absolute language	Most workloads gain nothing.
C	Camouflage	Budget does not change what quantum can do.
D	Correct	Narrow speedups, broad security impact.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	CAP trade-off	When a network partition occurs, a distributed system must choose between consistency (reject some requests) and availability (answer, possibly with stale data).	Believing a system can keep both during a partition.	A split between sites and a stated business priority.
2	Edge computing and latency	Signals in fiber travel about 200,000 km/s, so round-trip distance sets a latency floor; tight deadlines require processing near the source.	Counting one-way travel, or thinking a faster server beats distance.	A millisecond deadline and a distance to the data center.
3	Power usage effectiveness (PUE)	$PUE = \text{total facility power} \div \text{IT equipment power}$; 1.0 is ideal.	Inverting the ratio or using overhead alone.	Total power and IT power in kW.
4	Serverless computing	Serverless scales automatically, can scale to zero, and bills per use, which suits spiky or idle-heavy workloads; cold starts add latency.	Sizing fixed servers for peak all year.	Long quiet periods with brief large spikes.
5	Containers vs. virtual machines	Containers share the host kernel and start fast with low overhead; VMs run separate kernels and give stronger isolation.	Assuming containers isolate as strongly as VMs.	Density and speed needs vs. untrusted or mixed-OS workloads.
6	Zero trust	No request is trusted because of network location; every access is verified for identity, device, and least privilege.	Strengthening the perimeter while trusting everything inside.	One compromised inside device reaching many systems.
7	Domain-specific accelerators	With clock gains stalled, performance per watt now comes from specialized hardware (GPUs, TPUs, NPUs) matched to the workload.	Expecting higher CPU clock speeds to solve it.	One heavy workload repeated at huge scale with energy concerns.
8	Quantum computing in context	Quantum computers promise speedups for specific problem types, not all software, and threaten today's public-key cryptography, so organizations plan post-quantum migration.	Believing quantum makes everything faster or has no security impact.	Broad claims about quantum speed.

Course scope (public description): Emerging Computing Systems offers learners an exploration into the cutting-edge technologies and trends driving the future of computing, with a particular focus on internet computing systems. This course explores advanced computing topics, providing an in-depth analysis and evaluation of state-of-the-art technologies across nine broad categories: hardware architectures and operating systems, networking and distributed systems, cloud and data centers, system design and integration, security, performance and scalability, system integration, energy efficiency, and emerging technologies. The course will critically examine emerging technologies, their design principles, and their system optimizations and provide insights to the design and optimization of modern computing systems. This course provides a blend of conceptual

learning and practical application, allowing students to develop the expertise needed to lead in the evolving computing landscape. Computer Architecture and Systems is a prerequisite for this course.